

Number, units and estimation - Set 2 - Solutions

Specification M1.1, M1.2, M2.1-M2.14.

Answer key

Question	Answer	Topic
1	B	Number, units and estimation
2	D	Number, units and estimation
3	F	Number, units and estimation
4	A	Number, units and estimation
5	A	Number, units and estimation
6	A	Number, units and estimation
7	A	Number, units and estimation
8	H	Number, units and estimation
9	H	Number, units and estimation

Worked solutions

Question 1 NUM-16 · Number, units and estimation · spec M1.1

Water flows through a pipe at 1.2 litres per second. How long does it take to fill a tank of volume 3.6 m^3 ? ($1 \text{ m}^3 = 1000 \text{ litres}$.)

Answer: B

3600 litres at 1.2 litres/s takes $3000 \text{ s} = 50 \text{ minutes}$.

Full method - how and why

A rate problem with a unit conversion. Convert the volume to litres, divide by the rate, then convert the time.

- 1 Convert the volume: $3.6 \text{ m}^3 = 3.6 \times 1000 = 3600 \text{ litres}$.

why: The conversion is given in the question.

- 2 Divide by the rate: $\frac{3600}{1.2} = 3000 \text{ seconds}$.

why: Volume divided by volume-per-second gives seconds.

- 3 Convert to minutes: $\frac{3000}{60} = 50 \text{ minutes}$.

why: The options are in minutes, so finish the conversion.

Where it goes wrong

- Leaving the answer as 3000 seconds and picking 500 seconds by a slip.
- Dividing by 1.2 as if it were 12, giving 5 minutes.

Takeaway Make the units match before dividing, and convert the answer into the units the options use.

Question 2 NUM-17 · Number, units and estimation · spec M2.13

$a = 5.0$ and $b = 2.0$, each correct to 2 significant figures. Find the lower bound of $\frac{a}{b}$.

Answer: D

Smallest numerator over largest denominator: $\frac{4.95}{2.05}$.

Full method - how and why

Bounds for a quotient: smallest over largest gives the smallest result.

- 1 $a = 5.0$ to 2 s.f. means $4.95 \leq a < 5.05$.

why: The last significant figure is in the first decimal place, so the half-width is 0.05.

- 2 $b = 2.0$ to 2 s.f. means $1.95 \leq b < 2.05$.

why: Same reasoning.

- 3 For the **smallest** quotient, make the numerator as small as possible and the denominator as large as possible.

why: Increasing the denominator decreases a positive fraction - this is the key insight.

- 4 So the lower bound is $\frac{4.95}{2.05}$.

why: Left in fraction form, which is what the options offer.

Where it goes wrong

- Pairing both lower bounds as $\frac{4.95}{1.95}$: a smaller denominator makes the fraction **bigger**, so that is not the lower bound.
- Using ± 0.5 instead of ± 0.05 - for these numbers, 2 significant figures means one decimal place.

Takeaway Quotients: lower bound $= \frac{\text{lower}}{\text{upper}}$, upper bound $= \frac{\text{upper}}{\text{lower}}$.

Question 3 NUM-18 · Number, units and estimation · spec M2.3

What is the smallest positive integer that is divisible by every integer from 1 to 10?

Answer: F

$$\text{lcm}(1, \dots, 10) = 2^3 \cdot 3^2 \cdot 5 \cdot 7 = 2520.$$

Full method - how and why

The smallest number divisible by all of 1 to 10 is their LCM: take the highest power of each prime up to 10.

- 1 List the primes up to 10: 2, 3, 5, 7.

why: Only primes matter; composite numbers are built from them.

- 2 Take the highest power of each that is at most 10: $2^3 = 8$, $3^2 = 9$, 5, 7.

why: 8 needs three 2s and 9 needs two 3s - the largest demands from any single number.

- 3 Multiply: $8 \times 9 \times 5 \times 7 = 2520$.

why: $72 \times 35 = 2520$.

- 4 Check: 2520 is divisible by 4, 6, 10 automatically, since those are built from the primes included.

why: Covering the highest prime powers covers every composite too.

Where it goes wrong

- Multiplying all the numbers 1 to 10 (3628800) - divisible, but far from smallest.
- Using 2^2 instead of 2^3 , giving 1260, which is not divisible by 8.

Takeaway LCM of a range: each prime to the highest power that appears within the range.

Question 4 NUM-19 · Number, units and estimation · spec M2.9

Which of the following fractions has a terminating decimal expansion?

Answer: A

A fraction in lowest terms terminates iff its denominator has no prime factors other than 2 and 5: $40 = 2^3 \cdot 5$.

Full method - how and why

A fraction terminates exactly when its lowest-terms denominator is built only from 2s and 5s.

- 1 Check each denominator is in lowest terms with the numerator.

why: A shared factor could change the denominator - here none do.

- 2 Factorise: $40 = 2^3 \times 5$ ✓; $12 = 2^2 \times 3$ ✗; $15 = 3 \times 5$ ✗; $14 = 2 \times 7$ ✗; $9 = 3^2$ ✗; $22 = 2 \times 11$ ✗.

why: Any prime other than 2 or 5 forces a recurring expansion.

- 3 So only $\frac{7}{40}$ terminates (it is 0.175).

why: Because $10 = 2 \times 5$, only these primes can divide a power of ten.

Where it goes wrong

- Assuming a 'nice-looking' denominator like 12 terminates.
- Forgetting to reduce a fraction first in questions where the numerator shares a factor.

Takeaway Terminating $\Leftrightarrow$ denominator (in lowest terms) is $2^a 5^b$.

Question 5 NUM-20 · Number, units and estimation · spec M2.11

Find the exact value of $(\sqrt{5} - 1)^2 + (\sqrt{5} + 1)^2$.

Answer: A

$$(6 - 2\sqrt{5}) + (6 + 2\sqrt{5}) = 12.$$

Full method - how and why

A sum of conjugate squares. The surd terms cancel, so the answer is rational.

1 Expand the first: $(\sqrt{5} - 1)^2 = 5 - 2\sqrt{5} + 1 = 6 - 2\sqrt{5}$.
why: $(a - b)^2 = a^2 - 2ab + b^2$ with $(\sqrt{5})^2 = 5$.

2 Expand the second: $(\sqrt{5} + 1)^2 = 6 + 2\sqrt{5}$.
why: Only the middle sign differs.

3 Add: the $\mp 2\sqrt{5}$ terms cancel, leaving 12.
why: Conjugates always cancel their surd parts when added.

Where it goes wrong

- Dropping the cross terms: here it happens to give 12 anyway, but the same slip fails whenever the two brackets are not conjugates.
- Adding the surd terms instead of cancelling, giving $12 + 4\sqrt{5}$ - an offered option.

Takeaway $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$; here $2(5 + 1) = 12$, with no surds surviving.

Question 6 NUM-02 · Number, units and estimation · spec M2.3

The lowest common multiple of 12, 18 and n is 180. Which of the following could be n ?

Answer: A

$\text{lcm}(12, 18) = 36 = 2^2 \cdot 3^2$ and $180 = 2^2 \cdot 3^2 \cdot 5$, so n must be a divisor of 180 that is a multiple of 5. Only 20 qualifies: 24, 8 and 27 have no factor 5, and 50 and 40 do not divide 180.

Full method - how and why

An LCM condition. Work out what the other two numbers already supply, then see what n must add - and must not exceed.

1 Factorise: $12 = 2^2 \times 3$, $18 = 2 \times 3^2$, so $\text{lcm}(12, 18) = 2^2 \times 3^2 = 36$.
why: LCM takes the highest power of each prime.

2 The target is $180 = 2^2 \times 3^2 \times 5$, so n must supply the factor 5.
why: 36 has no factor 5, so n is the only possible source.

3 n must also **divide** 180: it cannot contain 2^3 , 3^3 or 5^2 , or the LCM would be larger than 180.
why: The LCM is at least n , so n must divide the target exactly.

4 Check the options: $20 = 2^2 \times 5$ divides 180 and supplies the 5 ✓. The others either lack a factor 5 (24, 8, 27) or do not divide 180 ($50 = 2 \times 5^2$, $40 = 2^3 \times 5$).
why: Test both conditions on each candidate.

Where it goes wrong

- Choosing 50 or 40 because they contain a 5, forgetting they must divide 180.
- Assuming n must equal $180 \div 36 = 5$.

Takeaway For $\text{lcm}(a, b, n) = L$: n must divide L and supply any prime powers a and b lack.

Question 7 NUM-09 · Number, units and estimation · spec M2.3

How many positive divisors does 360 have?

Answer: A

$360 = 2^3 \cdot 3^2 \cdot 5$, so the number of divisors is $(3 + 1)(2 + 1)(1 + 1) = 24$.

Full method - how and why

Counting divisors from the prime factorisation: each prime's exponent can be chosen independently.

1 Factorise: $360 = 8 \times 45 = 2^3 \times 3^2 \times 5$.

why: Split into convenient factors and factorise each.

2 A divisor has the form $2^a 3^b 5^c$ with $0 \leq a \leq 3, 0 \leq b \leq 2, 0 \leq c \leq 1$.

why: Any such combination divides 360, and every divisor has this form uniquely.

3 Count the choices: $4 \times 3 \times 2 = 24$.

why: Each exponent has (power + 1) possible values, because 0 is allowed.

Where it goes wrong

- Multiplying the exponents ($3 \times 2 \times 1 = 6$) instead of (exponent + 1).
- Forgetting that 1 and 360 are themselves divisors - the '+1's account for them.

Takeaway Number of divisors of $p^a q^b r^c$ is $(a + 1)(b + 1)(c + 1)$.

Question 8 NUM-12 · Number, units and estimation · spec M2.3

n is a positive integer such that $n, n + 2$ and $n + 4$ are all prime. Which of the following statements is/are true?

- I There is exactly one such value of n .
- II n must be odd.
- III One of $n, n + 2, n + 4$ must be divisible by 3.

Answer: H

Among any three numbers of the form $n, n + 2, n + 4$ one is a multiple of 3 (they cover all residues mod 3) - III. So one of them must equal 3 itself for all to be prime, giving $n = 3$ only (3, 5, 7) - I. And 3 is odd - II.

Full method - how and why

A prime-triple question. One observation about remainders on division by 3 settles all three statements.

1 III: among $n, n + 2, n + 4$, the remainders on division by 3 are all different (they differ by 2 and 4, i.e. by 2 and 1 modulo 3), so one of them is a multiple of 3. **True.**

why: Three numbers hitting all three remainder classes must include one divisible by 3.

2 I: that multiple of 3 can only be prime if it **equals** 3. Checking the possibilities gives $n = 3$ (3, 5, 7 are all prime); $n = 1$ fails (1 is not prime) and larger cases contain a composite multiple of 3. **True.**

why: So the triple is unique - this is why 3, 5, 7 is famous as the only 'prime triplet' of this form.

3 II: $n = 3$ is odd. (Also, any even $n > 2$ would not be prime.) **True.**

why: Both routes give the same verdict.

4 All three.

why: Match to the option list.

Where it goes wrong

- Testing a few values, finding only 3, 5, 7, and still doubting uniqueness - the mod-3 argument proves it.
- Thinking n could be 2: then 4 is not prime.

Takeaway Any three numbers spaced 2 apart cover all remainders mod 3, so one is divisible by 3.

Question 9 X2-18 · Number, units and estimation · spec M2.3

n is a positive integer greater than 1. Which of the following statements is/are true?

- I If n is not prime, then n has a prime factor at most $\sqrt{n}$.
- II If n has no prime factor at most $\sqrt{n}$, then n is prime.
- III If n is prime, then n has no factor between 1 and n exclusive.

Answer: H

If $n = ab$ with $1 < a \leq b$ then $a \leq \sqrt{n}$ and a has a prime factor $\leq \sqrt{n}$ (I). II is the contrapositive of I. III is the definition of prime.

Full method - how and why

Three number-theory statements about primality testing, two of which are contrapositives of each other.

- 1 I: if n is composite write $n = ab$ with $1 < a \leq b$. Then $a^2 \leq ab = n$, so $a \leq \sqrt{n}$.
why: The smaller factor of a pair cannot exceed the square root, or the product would be too big.
- 2 $a > 1$ has a prime factor, which is at most $a \leq \sqrt{n}$. So n has a prime factor $\leq \sqrt{n}$. **True.**
why: Every integer greater than 1 has a prime factor, and it cannot exceed the number itself.
- 3 II: 'no prime factor $\leq \sqrt{n} \Rightarrow n$ prime' is exactly the contrapositive of I, so it is **True** automatically.
why: Contrapositives share truth values - no separate work needed.
- 4 III: 'if n is prime it has no factor strictly between 1 and n ' is the definition of prime. **True.**
why: A prime's only positive factors are 1 and itself.
- 5 So all three.
why: Match to the list.

Where it goes wrong

- Treating II as the converse of I and rejecting it; it is the contrapositive (negate and swap).
- Wording III as 'no factors at all', which is wrong: 1 and n are factors.

Takeaway To test n for primality, trial-divide by primes up to $\sqrt{n}$ - and I/II are the two readings of why that works.